

Solutions

Numbering follows the lecture notes. For TYPE C exercises the idea is boxed; the rest are routine once the idea is in hand, which is what the types mean.

Lectures 1 and 2

§1.8* Entropy, and why the Gaussian maximizes it

1.13 A $H(q) = -q \log q - (1-q) \log(1-q)$: symmetric about $q = \frac{1}{2}$, zero at the endpoints, maximal at $\frac{1}{2}$ with value $\log 2$. Uniform on n points: $\log n$.

1.14 B By Jensen for the concave $\log$, $H(p) = \mathbb{E}[\log \frac{1}{p(X)}] \leq \log \mathbb{E}[\frac{1}{p(X)}] = \log \#\{x : p(x) > 0\} \leq \log n$. Equality in Jensen forces $1/p(X)$ a.s. constant, so p is uniform on its support; equality in the second step forces full support.

1.15 A $\log a$; $1 - \log \lambda$; $\frac{1}{2} \log(2\pi e \sigma^2)$. Uniform on $[0, a]$ with $a < 1$ has $h = \log a < 0$. Discrete entropy cannot be negative because $-p \log p \geq 0$ for $p \in (0, 1]$; a *density* may exceed 1, which is exactly what breaks that.

1.16 A $p_Y(y) = |a|^{-1} p_X(y/a)$; substituting $x = y/a$, $h(Y) = -\int p_X(x) [\log p_X(x) - \log |a|] dx = h(X) + \log |a|$.

1.17 B $-D(p \| q) = \int p \log \frac{q}{p} \leq \log \int q = 0$ by Jensen, with equality iff q/p is a.s. constant, hence $\equiv 1$. Under $X \mapsto aX$ both densities gain the same factor $|a|^{-1}$ and the same substitution is made, so $D(p \| q)$ is unchanged.

1.18 B $\log q(x) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma - \frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)$ is quadratic in x , so $\int p \log q$ is a fixed linear functional of $\int p$, $\int x_i p$, $\int x_i x_j p$. Matching mean and covariance therefore gives $\int p \log q = \int q \log q = -h(q)$, and $0 \leq D(p \| q) = -h(p) + h(q)$. For $h(q)$: $\mathbb{E}[(X - \mu)^\top \Sigma^{-1} (X - \mu)] = \text{tr}(\Sigma^{-1} \Sigma) = n$, so $h(q) = \frac{1}{2} \log((2\pi e)^n \det \Sigma)$.

1.19 C

$(x, y) \mapsto (\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}})$ is **orthogonal**: Jacobian 1, so it preserves joint differential entropy.

With $U = \frac{X+Y}{\sqrt{2}}$, $V = \frac{X-Y}{\sqrt{2}}$: $h(U, V) = h(X, Y) = 2h(X)$ by independence. Subadditivity $h(U, V) \leq h(U) + h(V)$ is Exercise 1.17 applied to the joint density against the product of its marginals. Symmetry gives $(X, Y) \stackrel{d}{=} (X, -Y)$, so $V \stackrel{d}{=} U$. Chaining, $2h(X) \leq 2h(U)$.

1.20 C

The Gaussian proof never used Gaussianity. It used that $\log q$ is an **affine combination of the constrained functions**, so $\int p \log q$ is fixed by the constraints alone.

$\log q = \sum_m \lambda_m T_m - \log Z$, so for any feasible p , $0 \leq D(p \| q) = -h(p) - \sum_m \lambda_m c_m + \log Z$ — and the last two terms do not depend on p . Taking $p = q$ evaluates them as $h(q)$; subtracting gives $h(p) \leq h(q)$, equality iff $p = q$. (No integrability of $h(p)$ need be assumed: $-\infty$ makes it trivial, $+\infty$ is excluded by $D(p \| q) \geq 0$.)

Instances: no constraint on $[0, a]$ gives q constant; $T_1(x) = x$ on $[0, \infty)$ gives the exponential; T 's $= x_i, x_i x_j$ give $\mathcal{N}(\mu, \Sigma)$.

Why the assumption is not free. On $S = \mathbb{R}$ with only the mean constrained, $e^{\lambda x}$ is never integrable, and correctly so: $\mathcal{N}(0, \sigma^2)$ is feasible for every σ with $h = \frac{1}{2} \log(2\pi e \sigma^2) \rightarrow \infty$. “ **q is a density**” is exactly what rules this out, and a student who assumes existence has assumed the content.

§2.5 Bayesian linear models, and §2.6*

2.9 A $p(y | w) = (2\pi\sigma^2)^{-n/2} e^{-\|y - Hw\|^2 / 2\sigma^2}$, so $-\log p = \frac{1}{2\sigma^2} \|y - Hw\|^2 + \frac{n}{2} \log(2\pi\sigma^2)$ and the minimizer over w is the least-squares fit. Choosing squared error is asserting that the noise is Gaussian, homoscedastic and independent across observations; most people choosing it are not asserting anything, which is the point.

2.10 A The posterior density is $\propto \exp(-\frac{1}{2\sigma^2} \|y - Hw\|^2 - \frac{1}{2\tau^2} \|w\|^2)$, so its maximizer minimizes $\|y - Hw\|^2 + \lambda \|w\|^2$ with $\lambda = \sigma^2 / \tau^2$. Large λ means noisy data against a confident prior, so heavy shrinkage; $\tau \rightarrow 0$ gives $\hat{w} = 0$ and $\tau \rightarrow \infty$ gives least squares.

2.11 B (w, y) is a linear image of (w, ξ) , hence jointly Gaussian, with $\Sigma_{ww} = \tau^2 I_p$, $\Sigma_{wy} = \mathbb{E}[w(Hw + \sigma\xi)^\top] = \tau^2 H^\top$, $\Sigma_{yy} = \tau^2 H H^\top + \sigma^2 I_n$. Corollary 2.5 then gives the two displays directly. $\text{Cov}(w | y)$ contains no y : the

data changes your *estimate*, never your *uncertainty*. Operationally, a surprising y moves the posterior mean and leaves the error bars exactly where they were — special to the Gaussian model, and false in general.

2.12 B $\text{Cov}(X, Y) = \mathbb{E}[(Y^2 - 1)Y] = \mathbb{E}Y^3 = 0$, so the best linear predictor is 0 with error $\text{Var } X = \text{Var}(Y^2) = 2$. The predictor $f(Y) = Y^2 - 1$ has error 0. What fails is **joint** Gaussianity of (X, Y) — X is not Gaussian at all — so Theorem 2.3 does not apply, and the conditional mean is not linear.

2.13 C

$H^\top(HH^\top + \lambda I_n) = (H^\top H + \lambda I_p)H^\top$ — one line of expansion. Multiply by the two inverses.

Both sides equal $H^\top HH^\top + \lambda H^\top$. Left-multiplying by $(H^\top H + \lambda I_p)^{-1}$ and right-multiplying by $(HH^\top + \lambda I_n)^{-1}$ gives the identity. The right-hand form inverts an $n \times n$ matrix, so the cost is $O(n^3)$; this is the kernel trick, and why Gaussian process regression is feasible.

As $\tau \rightarrow \infty$ we have $\lambda \rightarrow 0$, and in the form $H^\top(HH^\top + \lambda I_n)^{-1}y$ the limit is benign because $HH^\top$ is $n \times n$ and invertible: $\mathbb{E}[w | y] \rightarrow H^\top(HH^\top)^{-1}y$, the minimum-norm solution of $Hw = y$. In the other form the limit is $0 \cdot \infty$ — which is what makes this a C and not a computation.

Lectures 3 and 4

§1.7* The proof we did not give

1.9 A Differentiating $\varphi_X(t) = \mathbb{E}[e^{itX}]$ under the integral k times brings down $(iX)^k$. For $\varphi = e^{-t^2/2}$: $\varphi' = -t\varphi$, $\varphi'' = (t^2 - 1)\varphi$, so $\varphi''(0) = -1 = i^2 \mathbb{E}g^2$. The series $e^{-t^2/2} = 1 - \frac{t^2}{2} + \frac{t^4}{8} - \dots$ has t^4 coefficient $\frac{1}{8}$, so $\varphi^{(4)}(0) = 4!/8 = 3 = i^4 \mathbb{E}g^4$.

1.10 B With $\theta = tX$, $\mathbb{E}[e^{itX}] = 1 - \frac{t^2}{2} + r(t)$ where $|r(t)| \leq \mathbb{E} \min(|t|^3|X|^3/6, t^2X^2)$. Divide by t^2 : the integrand is bounded by X^2 , which is integrable, and tends to 0 pointwise, so dominated convergence gives $r(t)/t^2 \rightarrow 0$. The second branch of the minimum is what supplies the dominating function without a third moment.

1.11 B With $a = 1 + z_n/n$ and $b = e^{z_n/n}$, both are $1 + O(1/n)$ so $|a|, |b| \leq M$ for some fixed M , and $|a - b| = O(1/n^2)$; hence $|a^n - b^n| \leq nM^{n-1}|a - b| \rightarrow 0$ while $b^n = e^{z_n} \rightarrow e^z$. Applying this with $z_n = n(\varphi_{X_1}(t/\sqrt{n}) - 1) \rightarrow -t^2/2$ by Exercise 1.10 gives $\varphi_{S_n}(t) \rightarrow e^{-t^2/2}$, and Lévy finishes. **Independence** is what gives $\varphi_{S_n} = \prod_i \varphi_{X_i}(t/\sqrt{n})$; **identical distribution** only lets us write one factor n times.

1.12 A $\varphi_B(t) = (1 - p) + pe^{it} = 1 + p(e^{it} - 1)$. So $\varphi_{N_n} = [1 + \frac{\lambda}{n}(e^{it} - 1)]^n \rightarrow \exp(\lambda(e^{it} - 1))$ by Exercise 1.11, and $\sum_k e^{-\lambda} \frac{\lambda^k}{k!} e^{itk} = e^{-\lambda} e^{\lambda e^{it}}$ is the same thing.

1.13 B $X_{n,i}$ takes $1 - \lambda/n$ with probability λ/n and $-\lambda/n$ otherwise, so $s_n^2 = n \cdot \frac{\lambda}{n}(1 - \frac{\lambda}{n}) = \lambda(1 - \frac{\lambda}{n})$. For $\varepsilon < 1/\sqrt{\lambda}$ the threshold $\varepsilon s_n \rightarrow \varepsilon\sqrt{\lambda} < 1$ eventually falls below the large value, so $L_n(\varepsilon) \rightarrow n \cdot \frac{\lambda}{n} \cdot 1/\lambda = 1$.

At $\lambda = 3$, $\varepsilon = \frac{1}{2}$: $n = 10$ gives $s_n = 1.449$ and $\varepsilon s_n = 0.7246$, while the large value is $1 - \frac{3}{10} = 0.7 < 0.7246$ — nothing is caught, so $L_{10} = 0$ **exactly**. Then 0.970 and 1.000.

1.14 C

A Gaussian characteristic function $e^{-\sigma^2 t^2/2}$ **never vanishes**. So exhibiting one zero of the limit is a complete proof that the limit is not Gaussian.

$s_n^2 = n \cdot 1 + n = 2n$. The large summand has $|X_{n,n+1}| = \sqrt{n} > \varepsilon\sqrt{2n} = \varepsilon s_n$ exactly when $\varepsilon < 1/\sqrt{2}$, and it alone contributes $n/2n = \frac{1}{2}$ to L_n . The first n summands give $s_n^{-1} \sum_{i \leq n} X_{n,i} \Rightarrow \mathcal{N}(0, \frac{1}{2})$ with transform $e^{-t^2/4}$; the last gives $\pm 1/\sqrt{2}$ with transform $\cos(t/\sqrt{2})$; they are independent, so the product is the limit. It vanishes at $t = \pi/\sqrt{2}$.

1.15 C

Write $\mathbb{E}f(X_\Sigma) = \frac{1}{(2\pi)^n} \int \hat{f}(t) e^{-\frac{1}{2}t^\top \Sigma t} dt$, so that Σ appears **only in the exponent**.

That identity is Fourier inversion plus Definition 1.2. Differentiating in s under the integral brings down $-\frac{1}{2} \sum_{ij} \dot{\Sigma}_{ij} t_i t_j$. Since $\widehat{\partial_i \partial_j f}(t) = (it_i)(it_j)\hat{f}(t) = -t_i t_j \hat{f}(t)$, this is $\frac{1}{2} \sum_{ij} \dot{\Sigma}_{ij} \widehat{\partial_i \partial_j f}(t)$, and inverting the transform term by term gives the claim. Integrating over $\Sigma(s) = (1 - s)\Sigma^0 + s\Sigma^1$, for which $\dot{\Sigma} = \Sigma^1 - \Sigma^0$, gives the second display.

Through the density instead one is differentiating Σ^{-1} and $\det \Sigma$: the same answer, a page later.

Check. In $\mathbb{R}^2$ with unit variances and correlation ρ , $\frac{d}{d\rho} \mathbb{E}[X_1^2 X_2^2] = \mathbb{E}[4X_1 X_2] = 4\rho$, against Wick's $\frac{d}{d\rho}(1 + 2\rho^2) = 4\rho$.

1.16 C

S_n has an **atom**. A distribution function with a jump of size a is at least $a/2$ from any continuous one.

If $F(z_0) - F(z_0^-) = a$ and Φ is continuous then $|F(z_0) - \Phi(z_0)| + |F(z_0^-) - \Phi(z_0^-)| \geq a$, so the sup is $\geq a/2$. For $X_i = \pm 1$ and n even, $\mathbb{P}(S_n = 0) = \binom{n}{n/2} 2^{-n} \sim \sqrt{2/\pi n}$, giving $\sup_z |\mathbb{P}(S_n \leq z) - \Phi(z)| \geq c n^{-1/2}$. Lattice summands were slowest in §1.5 for exactly this reason: their atoms cannot be smoothed, and they are of the same order as the whole error budget.

§2.5* Where independence was spent

2.10 A $\mathbb{E}F(W, W') = \mathbb{E}F(W', W)$ by exchangeability, and $= -\mathbb{E}F(W, W')$ by antisymmetry; so it is zero.

2.11 B $F(x, y) = (y - x)(f(x) + f(y))$ is antisymmetric, so $\mathbb{E}[(W' - W)(f(W) + f(W'))] = 0$. Writing $f(W') = f(W) + (W' - W)f'(W) + R$ with $|R| \leq \frac{\|f''\|_\infty}{2}(W' - W)^2$,

$$0 = 2\mathbb{E}[(W' - W)f(W)] + \mathbb{E}[(W' - W)^2 f'(W)] + \mathbb{E}[(W' - W)R].$$

The first term is $-2a\mathbb{E}[Wf(W)]$ by the Stein-pair property and conditioning on W . Rearranging gives the display, with the remainder bounded by $\frac{\|f''\|_\infty}{4a}\mathbb{E}|W' - W|^3$.

2.12 B Swapping $X_I \leftrightarrow X'_I$ is measure preserving, so (W, W') is exchangeable. Since $\mathbb{E}[X'_I | X_1, \dots, X_n] = 0$ and $\mathbb{E}[X_I | X_1, \dots, X_n] = \frac{1}{n} \sum_i X_i$, $\mathbb{E}[W' - W | X_1, \dots, X_n] = -\frac{1}{\sqrt{n}} \cdot \frac{1}{n} \sum_i X_i = -W/n$, and the tower property gives $a = 1/n$. Also $(W' - W)^2 = (X_I - X'_I)^2/n$, whose conditional expectation is $\frac{1}{n^2} \sum_i (X_i^2 + 1)$. Dividing by $2a = 2/n$ gives $\frac{1}{2}(1 + \frac{1}{n} \sum_i X_i^2)$, whose distance from 1 is $\frac{1}{2}|1 - \frac{1}{n} \sum_i X_i^2|$, at most $\frac{1}{2}\sqrt{\text{Var}(X_1^2)/n}$ in L^1 . The remainder is $\frac{n\|f''\|_\infty}{4} n^{-3/2} \mathbb{E}|X_1 - X'_1|^3$. Both are $O(n^{-1/2})$.

Where independence entered: in the two conditional expectations, and nowhere else. The product formula $\mathbb{E}[X_i f(W^{(i)})] = \mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]$ never appeared.

2.13 A $\mathbb{E}W = n \cdot \frac{1}{n} = 1$; $\mathbb{E}[I_i I_j] = \frac{1}{n(n-1)}$ for $i \neq j$ gives $\mathbb{E}W^2 = 1 + 1 = 2$, so $\text{Var } W = 1$. And $\mathbb{P}(I_1 = I_2 = 1) = \frac{1}{n(n-1)} \neq \frac{1}{n^2}$. What fails is **independence of the** I_i , which Theorem 2.7 requires; the theorem therefore says nothing, and a true conclusion reached from unsatisfied hypotheses is still not proved.

2.14 C

Force i to be a fixed point by **composing with the transposition that sends i to itself**: $\sigma = \pi \circ (i \ \pi^{-1}(i))$.

Then $\sigma(i) = \pi(\pi^{-1}(i)) = i$, and the map is a bijection from $\{\pi : \pi(i) \neq i\}$ onto itself composed with a relabelling, so σ is uniform among permutations fixing i . Writing $j = \pi^{-1}(i)$: i becomes fixed (gain 1), and j becomes fixed exactly when $\pi(i) = j$, i.e. when π has a 2-cycle at i (gain 2). Hence $V_i = W(\sigma) - 1 \in \{W, W + 1\}$, and

$$\mathbb{E}|W - V_i| \leq \mathbb{P}(\pi(i) = i) + \mathbb{P}(\text{2-cycle at } i) = \frac{1}{n} + \frac{1}{n}.$$

Chen–Stein with $p_i = 1/n$, $\lambda = 1$ gives $d_{\text{TV}}(W, \text{Poisson}(1)) \leq n \cdot \frac{1}{n} \cdot \frac{2}{n} = \frac{2}{n}$.

At $n = 10$ the bound is 0.2 and the truth is 2.3×10^{-8} : wrong by about seven orders of magnitude. What it has that inclusion–exclusion does not is **robustness** — it never used that π was a permutation beyond the coupling, and it is the only one of the three routes in this course that produces a bound at all.

Lectures 5 and 6

§1 Wick's theorem

1.6 A The first point has $2q - 1$ choices of partner; deleting that block leaves $2q - 2$ points, so the count is $(2q - 1)!!$, giving 1, 3, 15, 105. Wick with $n = 1$, $\Sigma = 1$ makes every product of covariances equal 1, so $\mathbb{E}[g^{2q}]$ is that count; $\mathbb{E}[g^{20}] = 19!!$.

1.7 B $\mathbb{E}[X^\top A X] = \sum_{ij} A_{ij} \Sigma_{ij} = \text{tr}(A \Sigma)$. Wick gives $\mathbb{E}[X_i X_j X_k X_l] = \Sigma_{ij} \Sigma_{kl} + \Sigma_{ik} \Sigma_{jl} + \Sigma_{il} \Sigma_{jk}$, so $\mathbb{E}[(X^\top A X)^2] = (\text{tr } A \Sigma)^2 + 2 \text{tr}(A \Sigma A \Sigma)$, the last two pair partitions being equal by symmetry of A and Σ . Subtracting the squared mean, $\text{Var}(X^\top A X) = 2 \text{tr}((A \Sigma)^2)$; with $A = I$, $\text{Var} \|X\|^2 = 2 \text{tr } \Sigma^2$.

§2.9* Counting the walks

2.12 A $\frac{1}{N} \mathbb{E} \operatorname{tr} M^2 = \frac{1}{N} \sum_{i,j} \mathbb{E} M_{ij}^2 = \frac{1}{N} [(N^2 - N) \frac{1}{N} + N \frac{2}{N}] = 1 + \frac{1}{N}$. Odd moments vanish with no computation: $\operatorname{tr} M^{2k+1}$ is a sum of products of an odd number of centred Gaussians, and Wick pairs factors two at a time.

2.13 B The r th factor of a term is the matrix entry of the r th step of the closed walk $i_1 \rightarrow \cdots \rightarrow i_{2k} \rightarrow i_1$. Since $\mathbb{E}[M_{ab}M_{cd}] = 0$ unless $\{a, b\} = \{c, d\}$, a block joining two steps on different edges kills the term. So the $2k$ steps are matched into k pairs, each pair on one edge: at most k distinct edges, each used at least twice, hence exactly twice in the leading terms.

2.14 B A closed walk has a connected edge graph, and a connected graph on v vertices has $e \geq v - 1$; with $e \leq k$ this gives $v \leq k + 1$, equality forcing $e = k$ and $e = v - 1$, i.e. a tree with k edges each traversed twice. Counting: $\sim N^v$ index tuples, k covariances of size $1/N$, and a $1/N$ in front, so the contribution is N^{v-k-1} , which vanishes unless $v = k + 1$.

2.15 C

Condition on the block containing the point 1. Non-crossing forces the interval it spans to be self-contained, splitting the problem into two **independent** problems of the same kind.

If that block is $\{1, b\}$, a block joining a point inside $(1, b)$ to one outside would cross it, so the $b - 2$ interior points pair among themselves and $b = 2j + 2$ is even; the rest splits into $2j$ inside and $2(n - 1 - j)$ outside. Hence $C_n = \sum_{j < n} C_j C_{n-1-j}$, $C_0 = 1$: 1, 1, 2, 5, 14, 42. Multiplying by w^n and summing turns the convolution into a product, $C - 1 = wC^2$; the root finite at $w = 0$ is $C(w) = \frac{1 - \sqrt{1 - 4w}}{2w}$, whose coefficients are $\frac{1}{n+1} \binom{2n}{n}$.

2.16 C

Forget where the walk is. Record only whether each step goes somewhere **new** (+1) or comes back (−1).

On a tree each edge is traversed twice, and a first traversal necessarily lands on a vertex not yet visited, since a tree has no cycles. So +1 marks first traversals, the partial sum after r steps is the number of edges opened but not closed — the current depth, hence ≥ 0 — and the walk is closed, so the total is 0. That is a Dyck path. Reading a Dyck path left to right and opening a new edge at each +1 reconstructs the tree and the traversal, so the map is a bijection, and there are C_k shapes by Exercise 2.15.

A contributing walk is a shape plus an injective labelling of its $k + 1$ vertices, so the count is exactly $C_k N(N - 1) \cdots (N - k)$. Multiplying by N^{-k} and N^{-1} gives $C_k(1 + O(1/N))$.

2.17 C

The two δ 's in the real covariance come from $M_{ij} = M_{ji}$ — there are **two ways** to glue a pair of entries. Hermitian symmetry leaves one.

With $\mathbb{E}[H_{ab}H_{cd}] = \frac{1}{N} \delta_{ad} \delta_{bc}$, the first block of the crossing pair partition forces $i_1 = i_4$ and $i_2 = i_3$, and the second forces $i_2 = i_1$ and $i_3 = i_4$. All four indices coincide, so $v = 1$ and the contribution is $N^{1-3} = N^{-2}$, against N^{-1} in the real case. The two non-crossing pair partitions are unaffected and still contribute 1 each.

2.18 A The histogram matches ρ_{sc} visibly by $N = 2000$, and the $\pm 1/\sqrt{N}$ picture is the same: the limit depends on the entries only through their variance, although Wick is false for them.